A Classical Reasoning Agent for Knowledge-Base Entailment: Forward Chaining, Resolution and Causal Propagation

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***Abstract*—We here present an end-to-end system that here decides whether a query here is entailed by the knowledge base (KB) of facts and rules, returning True or False. A deterministic or maybe called as rule-based front-end converts natural-language, first-order-logic (FOL) or causal text into one internal representation, and the final decision is made thus only by classical engines implemented from the scratch: semi-naive forward chaining with negation as failure, resolution refutation with unification, and the causal propagation over a structural causal model.Here no generative model is used at any point. On the public data loaded from Hugging Face the system thus reaches 99.92% on 5000 RuleTaker test queries (3.65 s in total), 94.60% on 500 CounterBench queries and 84.24% on the FOLIO validation split , hence , where a constant "False" baseline scores 64.5%. We report the error analyses and the known limitations.**

***Index Terms*—automated reasoning, forward chaining, resolution, unification, RuleTaker, FOLIO, CounterBench, counterfactual reasoning**

I. Introduction

Given a KB made of facts, rules and a query, the task is thus to decide whether the query is logically entailed by the KB or not. The assignment thus requires the pipeline *natural language* $\rightarrow$ *internal representation* $\rightarrow$ *classical reasoning* $\rightarrow$ *True/False* and forbids the use of a generative model to produce the answer. Our system thus follows this pipeline. Three public benchmarks guided development: RuleTaker [1] (English facts and rules with chains of increasing depth), FOLIO [2] (human-written stories with FOL annotations) and CounterBench [3] (counterfactual questions over causal graphs). All three are thus loaded from the Hugging Face Hub (`hitachi-nlp/ruletaker`, `yale-nlp/FOLIO`, `CounterBench/CounterBench`).

II. Representation

A. Terms, literals, rules and clauses

A constant is a string, a variable is a string starting with `?`, and a compound (Skolem) term is a pair `(name, args)`. A *literal* is a triple $(s, p, \mathbf{t})$ of a sign, a predicate and an argument tuple; e.g. "Bob is not kind" becomes $(\text{False}, \text{kind}, (\text{bob}))$ and "the cat chases the dog" becomes $\text{chase}(\text{cat}, \text{dog})$. A *rule* is a pair (B, h) with a body B (a list of literals, possibly negated) and a head literal h; facts are ground literals. For full FOL, formulas over $\forall, \exists, \neg, \wedge, \vee, \rightarrow, \leftrightarrow, \oplus$ and equality are converted to negation normal form, Skolemised and put into CNF, so a clause is a tuple of literals.

B. Structural causal model

For counterfactual text, "A causes B", "A and B together cause C", "A or B causes C" and "not A causes B" become Boolean equations over a DAG, e.g. Skul = Quaz $\wedge$ Plim.

C. Front-ends and router

Four deterministic front-ends exist: (i) an English parser for RuleTaker sentences (facts, "If … then …", "Big, red things are round", negation, conjunction, relations); (ii) parsers for RuleTaker's prefix and ProbLog forms; (iii) a recursive-descent FOL parser for FOLIO; (iv) a parser for CounterBench's causal text. A router inspects the input and selects the front-end and the engine.

III. Reasoning Algorithms

A. Forward chaining

The KB is saturated by semi-naive, predicate-indexed forward chaining [4]: a newly derived fact triggers only rules whose body mentions its predicate, and the remaining body literals are joined by unification against an index of known facts. The least fixpoint is

$$F_{k+1} = F_k \cup \{\, h\theta \;:\; (B, h) \in R,\; B\theta \subseteq F_k \,\}. \qquad (1)$$

Each derived fact stores the rule instance that produced it, so proof trees are available without extra work. Negated body literals use *negation as failure* (closed-world reading), evaluated after the monotone part has saturated; an open-world mode, which accepts only explicit negative facts, is also implemented for comparison.

B. Unification and resolution

Unification uses an occurs check over variables, constants and compound terms. For general FOL, the negated query is

TABLE I
PUBLIC-SET RESULTS (COLAB CPU).

Dataset / setting	Queries	Accuracy	Time (s)
RuleTaker dev, closed-world	3000	100.00%	≈1.8
RuleTaker dev, open-world	3000	74.93%	≈1.7
RuleTaker test	5000	99.92%	3.65
CounterBench	500	94.60%	0.12
FOLIO validation	203	84.24%	25.0

added to the clause set and a given-clause loop searches for the empty clause using the binary resolution rule [5]

$$\frac{C_1 \vee L_1 \qquad C_2 \vee \neg L_2 \qquad \theta = \mathrm{mgu}(L_1, L_2)}{(C_1 \vee C_2)\theta}. \tag{2}$$

The search is best-first: a priority queue orders clauses by weight (literal count plus term size), so short clauses are resolved first (set-of-support). Factoring, tautology and duplicate removal, clause-size limits and a wall-clock budget (1 s per query in our FOLIO runs) bound the search. If the clauses are Horn and the goal is a ground atom, the router uses forward chaining instead.

C. Causal propagation

Following Pearl's structural causal models [6], an intervention $\mathrm{do}(X{=}x)$ overrides the equation of X; the target is evaluated by depth-first recursion over the DAG with memoisation. Variables without incoming edges take their observed value, or default to true.

D. Robustness for noisy FOL

FOLIO contains malformed annotations, so: unbalanced parentheses are repaired heuristically (append missing ")"; for a surplus ")" open a "(" after the leading quantifier prefix); a premise that still fails to parse, or whose CNF exceeds 20,000 clauses ($\oplus$ and $\leftrightarrow$ blow up exponentially), is skipped, which can only make an entailment harder to prove and never creates a false one; equality uses reflexivity and symmetry axioms.

IV. EXPERIMENTS

A. Setup

Accuracy is the fraction of queries whose predicted label equals the gold label. Time is the total engine time over the whole set (parsing included) on a Google Colab CPU runtime. Samples are random with a fixed seed. For FOLIO the three-way label is collapsed to binary: only "True" counts as entailed.

B. Results

RuleTaker. In this release the labels follow the closed-world assumption, so negation as failure is the right reading: it scores 100% on every configuration of the 3000-query dev sample (NatLang, depth-0, 1, 2, 3, 3ext, 3ext-NatLang and 5), against 74.93% for the open-world reading (Table I). On the public test split the four errors out of 5000 lie in depth-3 and depth-3ext configurations (1+2+1).

TABLE II
COUNTERBENCH ACCURACY BY QUESTION TYPE (500 QUERIES).

Type	basic	joint	nested	cond.	backdoor
Queries	105	110	95	97	93
Accuracy	100%	95.45%	97.89%	92.78%	86.02%

CounterBench. Accuracy by question type is given in Table II.

FOLIO. The system scores 84.24% (171/203) with 201 queries answered by resolution and 2 by forward chaining. Always answering "False" gives 64.5% (131/203), since 62 labels are False and 69 Uncertain; the system is therefore about 20 points above that baseline. Before the robustness measures of Section III-D, 85 of 203 examples crashed in the parser and accuracy was 79.8%.

C. Error analysis and limitations

- RuleTaker: the printed failures are depth-3 stories with negated rule bodies; we have not fully diagnosed them.
- CounterBench: errors concentrate in the conditional and backdoor types, which contain observations ("We observed ..."). The engine currently uses an observation only for variables without incoming edges, so evidence about derived variables is not propagated back to their causes (abduction is missing).
- FOLIO: the annotations contain known errors, so some mismatches come from the data; equality has no term-substitution axioms; $\oplus$ chains longer than six terms are skipped; "Uncertain" is scored as False because the task is binary.

V. REASONING TRACES

The traces below are produced by the system's `solve` function; the indentation shows which rule derived each fact.

A. Forward chaining (assignment example).

```
Query: Bob is successful. -> True [forward-chaining]
successful(bob) [rule 1]
  smart(bob) [rule 0]
    young(bob) [given]
  kind(bob) [given]
```

B. RuleTaker dev example, depth 2.

```
Query: Bob is white. -> True [forward-chaining]
white(bob) [rule 5]
  green(bob) [given]
  red(bob) [rule 4]
    quiet(bob) [given]
```

C. Negation as failure.

```
Query: Bob is old. -> True [forward-chaining]
old(bob) [rule 0]
  not young(bob) [not derived, holds by NAF]
```

D. FOL with an existential goal (resolution).

```
Premises: forall x (Dog(x) -> Animal(x))
        forall x (Animal(x) -> exists y Eats(x,y))
        Dog(rex)
Query: exists y Eats(rex,y) -> True [resolution]
refutation found after 3 clauses
```

E. CounterBench, intervention do(not Fizo).

```
Query: Would Jext occur if not Fizo instead of Fizo?
do(not Fizo)
Blorn = NOT (Fizo) = 1 Plim = Blorn = 1
Quaz = Plim = 1 Skul = Quaz AND Plim = 1
Triv = Skul = 1 Yex = Triv = 1
Jext = Yex AND Skul = 1 -> True
```

VI. Conclusion

A small set of classical engines behind rule-based parsers solves RuleTaker almost perfectly, answers 94.6% of CounterBench questions and 84.2% of FOLIO validation questions within a short per-query budget. No generative model produces any answer. The clearest next improvements are abduction for observed derived variables in CounterBench, term-substitution axioms for equality, and a diagnosis of the remaining negated-body failures in RuleTaker.

References

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